

Does the Conservative Formula still work?

Replication design and extensions

MTH 9897 / Project 8

First-draft presentation

Synthetic demonstration results

Research question

Does combining stable stocks with momentum and shareholder payouts improve performance?

Comparisons

Market benchmark
Single-signal strategies
Results after trading costs

Current status

Executable research prototype
Historical equity data pending
Empirical conclusions remain open

Why combine these signals?

The combination targets weaknesses of a single screen

Component	What it favors	Potential weakness
Low volatility	Stocks with more stable returns	Defensive stocks can become expensive
Momentum	Stocks with strong recent price trends	Trends can reverse sharply
Net payout yield	Cash distributions and fewer shares	Payouts may be temporary or debt-funded

Research hypothesis: the combined ranks improve the risk-return tradeoff.

The portfolio selection rule

Paper recipe at each quarter end, with sufficient eligible history

1,000

Largest eligible US stocks by market capitalization

500

Lower-volatility half using 36 months of returns

100

Best average ranks on momentum and payout yield

Equal weights at formation. Quarterly rebalancing. Historical sample counts may be smaller.

Signal definitions in the draft

Signal	Draft construction	Preference
Volatility	36-month return standard deviation	Lower
Momentum	11-month price return, skipping latest month	Higher
Payout proxy	Dividend yield + 1 – shares / 24-month mean	Higher

Payout definition remains subject to reconciliation with the paper.

A worked payout example

Illustrative inputs, using the notebook's draft definition

Input	Example value
Trailing 12-month dividends per share	\$2
Current split-adjusted price	\$50
Current / 24-month mean adjusted shares	96 million / 100 million

Dividend yield = $2 / 50 = 4\%$

Payout proxy = $4\% + (1 - 96 / 100) = 8\%$

The share-count term measures net issuance indirectly. A split must never look like a buyback.

Information timing

Example: portfolio formed at December month end

Input or action	Window / timing
Volatility	36 months ending in December
Momentum	January through November
Selection	December month-end information
First earned return	Following January

Quarter-end execution is idealized. Late-reported inputs require an additional lag.

How the stock ranks determine selection

Illustration: four stocks after the volatility screen, two places available

Stock	Momentum rank	Payout rank	Average rank
A	3	2	2.5
B	1	3	2.0
C	2	1	1.5
D	4	4	4.0

Select C and B. Each receives 50% in this small example.

In the actual baseline, the best 100 names each start at 1%. Smaller rank values are better.

Historical study and demonstration data

Dimension	Intended empirical study	Current demonstration
Source	WRDS / CRSP	Seeded synthetic generator
Return period	1929 to latest full year	January 2003–December 2025
Universe	Historical eligible US stocks	1,100 artificial securities
Interpretation	Test the paper's findings	Illustrate the research workflow

Historical work must retain delisted names and reconcile legacy versus CIZ return conventions.

Data errors can create apparent performance

Risk	Required control
Survivorship bias	Include inactive and delisted securities
Future information	Use classifications and inputs known at formation
Corporate actions	Keep prices, dividends, and shares on a consistent basis
Missing months	Require consecutive observations in rolling windows
Terminal returns	Reconcile exit data and avoid double counting

Current draft: stop on unexplained missing holding returns and report formation sample sizes.

Portfolio accounting

Quarterly targets

Weights drift between rebalances

Trading costs

Charge both purchases and sales

Terminal events

Final proceeds move to cash

Missing held returns

Stop and reconcile the data

Costs, cash returns, and initial purchases enter the backtest explicitly.

A worked transaction-cost example

Illustration: rebalance 20% out of old positions and 20% into new positions

Total traded notional

20% sold + 20% bought = 40% of NAV

Cost at 10 bps per dollar traded

$0.40 \times 0.001 = 0.0004 = 4 \text{ bps of NAV}$

If gross monthly return is 2%:

Net return = $(1 - 0.0004) \times 1.02 - 1 = 1.9592\%$

The notebook counts both trade directions and includes the initial purchase.

Experiments

Comparison	Research purpose
Remove momentum or payout	Measure each signal's incremental contribution
Remove the volatility screen	Test whether defensive selection matters
Equal / cap-weight universe	Separate selection effects from weighting
Later periods and costs	Assess stability and implementation sensitivity

Performance across time

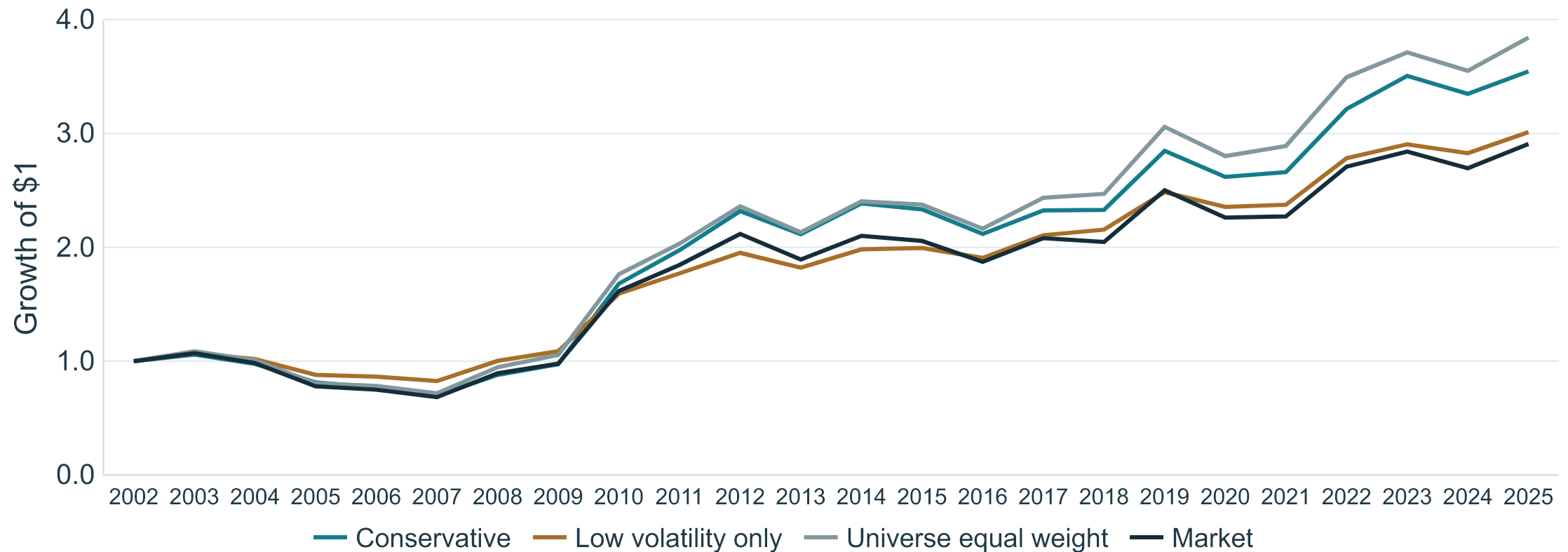
Period	Purpose	Interpretation
Through 2016	Compare with the paper	Match coverage and conventions first
2017–2018	Separate the transition	Two years is a short sample
2019 onward	Evaluate later performance	Keep portfolio rules unchanged

Report actual dates and observation counts for every comparison.

Later historical data are not an untouched prospective holdout for a researcher working in 2026.

Demonstration wealth paths

SYNTHETIC DEMONSTRATION



Year-end wealth, 2002–2025. Strategies include 10 bps per dollar traded. Synthetic market is gross.

How to read the performance measures

Measure	Meaning	Question it answers
CAGR	Compounded annual growth	How fast did wealth grow?
Volatility	Annualized monthly variability	How variable were returns?
Sharpe ratio	Excess return per unit of variability	Was return attractive relative to risk?
Maximum drawdown	Worst loss from a prior wealth peak	How severe was the deepest decline?
Turnover	Trading relative to portfolio size	How much implementation activity?

The market benchmark is gross. Strategy results include the chosen cost scenario.

Demonstration performance measures

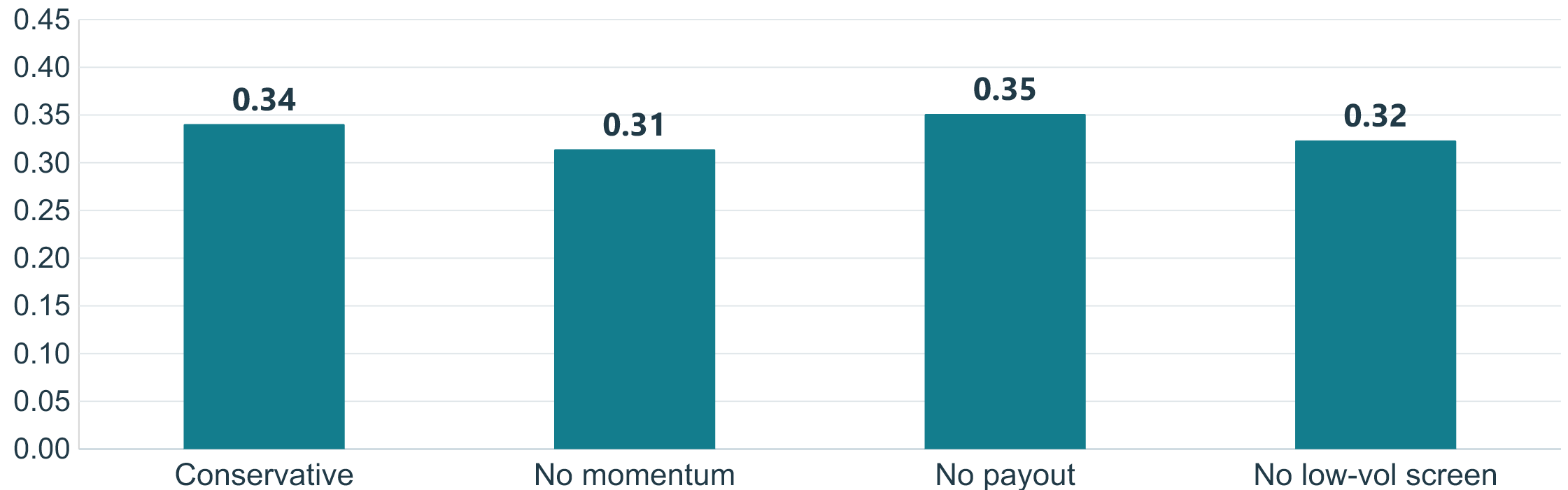
SYNTHETIC DEMONSTRATION

Portfolio	CAGR	Volatility	Sharpe	Max drawdown
Conservative	5.66%	13.61%	0.34	-38.28%
Low volatility only	4.91%	9.99%	0.35	-28.22%
Universe equal weight	6.03%	14.08%	0.36	-38.84%
Market	4.75%	14.16%	0.27	-41.06%

January 2003–December 2025. Synthetic results provide no evidence of real-world outperformance.

Synthetic signal comparisons

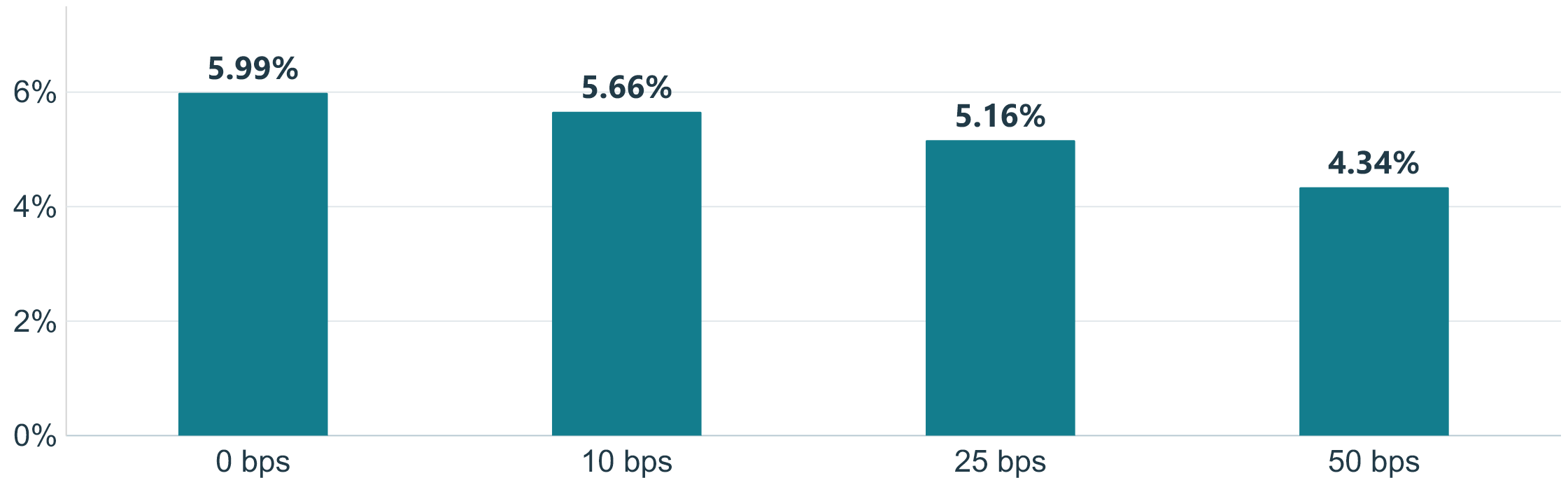
SYNTHETIC DEMONSTRATION



Annualized Sharpe, January 2003–December 2025. Interpretation awaits historical equity data.

Synthetic trading-cost sensitivity

SYNTHETIC DEMONSTRATION



Synthetic CAGR, 2003–2025. Basis points apply to each dollar traded. These are assumed cost scenarios.

What the demonstration illustrates

SYNTHETIC DEMONSTRATION

Comparison	Synthetic result	Interpretation
Growth vs low volatility	5.66% vs 4.91% CAGR	Higher growth in this simulation
Risk-adjusted performance	0.34 vs 0.35 Sharpe	The full formula is not uniformly better
Worst decline	-38.28% vs -28.22%	Low volatility has a smaller drawdown
Cost sensitivity	5.99% to 4.34% CAGR	0 to 50 bps assumed cost per dollar

All comparisons use 2003–2025 synthetic returns. They do not validate or refute the paper.

Empirical conclusions remain open

The draft defines a reproducible test of the Conservative Formula

Remaining work

Historical CRSP data and corporate-action audit
Payout definition and sample reconciliation
Empirical comparisons and uncertainty estimates

A successful replication may confirm, weaken, or reject the original findings.

References and project files

Van Vliet and Blitz (2018)

The Conservative Formula: Quantitative Investing Made Easy
SSRN 3145152

Supporting sources

Robeco, Guide to Conservative Investing (2023)
CRSP, Summary of CIZ Differences to Legacy Files
MTH 9897, Final Projects 2026

Notebook, slides, and generator code

github.com/yingnichang/MTH9897/tree/main/Final